

Worst Game of Blood on the Clocktower

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We develop an optimization model that maximizes the number of Jinxes in a specified-size game of Blood on the Clocktower. The game contains a set of characters, denoted C , indexed by i and j . C has pairwise disjoint subsets A , B , C , and D for the different characters types. We define the following parameters.

p_{ij}	-	boolean for if the character pair (i, j) forms a jinx
m_A	-	minimum number of characters from set A
M_A	-	maximum number of characters from set A
m_B	-	minimum number of characters from set B
M_B	-	maximum number of characters from set B
m_C	-	minimum number of characters from set C
M_C	-	maximum number of characters from set C
m_D	-	minimum number of characters from set D
M_D	-	maximum number of characters from set D
m_T	-	minimum number of characters overall
M_T	-	maximum number of characters overall

We let v_i be a boolean decision variable that is 1 if character i is in the optimal subset, 0 otherwise. Likewise we let e_{ij} be a boolean decision variable that is 1 if both (i, j) are in the optimal subset, forming a jinx. We then solve the following model

$$\begin{aligned} & \max \quad \frac{1}{2} \sum_{ij} p_{ij} e_{ij} \\ & \text{s.t.} \\ & \quad 2e_{ij} \leq v_i + v_j, \quad \forall i \forall j \\ & \quad \sum_{i \in A} v_i \geq m_A \\ & \quad \sum_{i \in A} v_i \leq M_A \\ & \quad \sum_{i \in B} v_i \geq m_B \\ & \quad \sum_{i \in B} v_i \leq M_B \\ & \quad \sum_{i \in C} v_i \geq m_C \\ & \quad \sum_{i \in C} v_i \leq M_C \\ & \quad \sum_{i \in D} v_i \geq m_D \\ & \quad \sum_{i \in D} v_i \leq M_D \\ & \quad \sum_i v_i \geq m_T \\ & \quad \sum_i v_i \leq M_T \end{aligned}$$

We create an instance of the model with $C = \{1, 2, \dots, 138\}$ and the following subsets.

$$A = \{1, \dots, 19\}, \quad B = \{20, \dots, 46\}, \quad C = \{47, \dots, 69\}, \quad \text{and} \quad D = \{70, \dots, 138\}.$$

The max and min parameters are given by

m_A	M_A	m_B	M_B	m_C	M_C	m_D	M_D	m_T	M_T
1	4	4	5	4	5	13	13	22	27

p_{ij} is set to one if the pair (i, j) or (j, i) appears in the set P below.

$P = \{(1, 32), (1, 38), (1, 121), (2, 38), (4, 31), (4, 40), (4, 78), (5, 40), (5, 53), (5, 62), (5, 69), (5, 88), (5, 108), (5, 111), (5, 120), (6, 35), (6, 53), (6, 77), (6, 89), (6, 90), (6, 98), (6, 101), (6, 104), (6, 110), (6, 112), (6, 123), (6, 124), (6, 131), (7, 31), (7, 37), (7, 38), (7, 41), (7, 53), (7, 108), (7, 119), (8, 32), (8, 54), (8, 129), (9, 40), (13, 40), (14, 75), (14, 77), (14, 89), (14, 90), (14, 98), (14, 101), (14, 104), (14, 110), (14, 112), (14, 123), (14, 124), (14, 131), (16, 32), (17, 77), (18, 89), (19, 40), (21, 54), (21, 61), (22, 50), (22, 52), (22, 54), (22, 60), (22, 62), (22, 71), (22, 85), (22, 136), (23, 61), (24, 28), (26, 61), (27, 41), (27, 61), (28, 61), (29, 54), (31, 40), (31, 61), (31, 71), (31, 76), (31, 100), (31, 108), (31, 109), (32, 71), (34, 48), (34, 71), (35, 40), (35, 49), (35, 52), (35, 54), (35, 60), (35, 62), (35, 85), (35, 136), (38, 61), (39, 49), (39, 54), (39, 60), (39, 61), (39, 71), (39, 108), (39, 119), (40, 53), (40, 71), (40, 83), (40, 84), (40, 88), (40, 119), (40, 120), (41, 62), (41, 69), (41, 72), (41, 84), (41, 102), (41, 108), (41, 120), (42, 49), (42, 54), (42, 71), (42, 108), (42, 119), (45, 61), (45, 71), (45, 108), (48, 79), (50, 109), (57, 109), (60, 64), (64, 124), (69, 79), (78, 117), (79, 103), (79, 121), (80, 109)\}.$

An optimal solution to this instance has 52 jinxes. The following v_i are set to one in this optimal solution,

5, 6, 7, 14, 35, 39, 40, 41, 42, 49, 53, 54, 62, 69, 71, 84, 88, 90, 98, 104, 108, 112, 119, 120, 123, 124, 131.

Converting numbers to names, this gives us Legion, Leviathan, Lil' Monsta, Riot, Pit-Hag, Spy, Summoner, Vizier, Widow, Damsel, Hatter, Heretic, Politician, Zealot, Alchemist, Courtier, Engineer, Farmer, Grandmother, King, Magician, Monk, Poppy Grower, Preacher, Ravenkeeper, Sage, Soldier,